# RP3 Rate and Temperature – Knowledge Organiser

*AQA A-level Chemistry • Set 1: Rate and Temperature* **Paper 2**

| # | Question | Answer |
|---|---|---|
| 1 | Reactants used | Sodium thiosulfate and hydrochloric acid |
| 2 | Equation | $Na_2S_2O_3 + 2HCl \rightarrow 2NaCl + SO_2 + S + H_2O$ |
| 3 | Precipitate formed | Sulfur |
| 4 | What is timed | Time for the cross to disappear |
| 5 | Rate is proportional to | 1/time |
| 6 | Independent variable | Temperature |
| 7 | Dependent variable | Time for the cross to disappear |
| 8 | Two control variables | Concentrations and volumes |
| 9 | How temperature is controlled | Water bath |
| 10 | Main hazard | $SO_2$ is toxic |